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Societal benefits and environmental performance of Chinese aquaculture

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ABSTRACT

Aquaculture has emerged as an alternative food production system, but it comes with both benefits and costs that need to be balanced. To evaluate the sustainable status of Chinese aquaculture, we quantify the benefits and costs of aquaculture in each province in China. We obtained the data on aquaculture production from the China Fisheries Statistical Yearbooks (2021) and used nine indicators to represent aquaculture benefits (nutrient, economy, and employment) and aquaculture costs (freshwater, energy, and land use, greenhouse gas, nitrogen, and phosphorus emissions). We found that the nine indicators varied among the 30 provinces in this study. The Southwest and South regions showed the highest benefits ($SBS = 30.65$) and costs ($EPS = 38.55$) at the regional level in composite aquaculture (freshwater and mariculture) respectively. At the provincial level, Tibet and Hainan had the largest score for aquaculture benefits ($SBS = 82.90$) and costs ($EPS = 61.37$) in composite aquaculture, respectively. Tibet province performed the best in freshwater and composite aquaculture (6.09 and 4.86 respectively), while Jiangsu province performed the best in mariculture according to benefits-to-costs ratio (2.71). More sustainable species production and use cleaner energy (such as hydro energy and nuclear energy etc) and reduce the feed conversion ratio are the direction of more sustainable aquaculture. This study provides the first analysis of Chinese aquaculture sustainability by benefits and costs, and more accurate data and more indicators through weighting could be included in this framework to achieve greater aquaculture assessment in the future.

1. Introduction

Aquaculture is an increasingly important sector as it provides an alternative source to traditional food production systems, such as capture fisheries and terrestrial livestock (Cao et al., 2011). Aquaculture continues to dominate aquatic food production globally, as wild fish production is declining to meet human demands (FAO, 2020; Tacon, 2020). This has led to aquaculture becoming a significant source of nutrition supply for ensuring human food security, surpassing global capture fisheries production (Clavelle et al., 2019; Tacon, 2020). Over the past few decades, global aquaculture production has increased at an average annual rate of 6%, rising from 34.2 million tonnes (Mt, 10^9 kg) in 1997 to 122.6 Mt in 2020 (FAO, 2020; FAO, 2022). While aquaculture production has experienced rapid expansion, significant disparities persist in the distribution of aquatic product output among various regions and nations. Notably, the Asian region, spearheaded by China, stands at the forefront of aquaculture, contributing to over 90% of the worldwide aquaculture yield (FAO, 2022). Furthermore, aquaculture has great potential to increase further, given the growing demand for animal protein in the future. Apart from providing protein, aquaculture

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is also a valuable source of micro-nutrients, such as iron, zinc, calcium, and iodine, which are deficient in many countries (Golden et al., 2021). The economic value of aquaculture is also crucial for people in rural areas. According to Filipski and Belton (2018), aquaculture generates higher incomes per-acre than agriculture, and it has larger income spillovers than agriculture for non-farm households through retail and labor markets. Aquaculture, therefore, plays an important role in rural development and poverty reduction. The range of cultivated species has seen substantial growth, with notable contributions from seaweeds, carps, bivalves, tilapia, and catfish, accounting for a significant 75% of aquaculture production (Naylor et al., 2021). Molluscs and algae, categorized as extractive species, distinguish themselves from other cultivated varieties by not requiring feed, rendering them appealing options for the expansion of sustainable seafood (Chopin and Tacon, 2021; Naylor et al., 2021). Finally, the aquaculture industry provides significant employment opportunities. Valderrama et al. (2010) estimated that the world aquaculture industry generated a total of 23.4 million jobs in 2005, with the Far East plus India accounting for about 92% of the total employment.

Aquaculture, as the fastest growing food production industry, can also pose environmental challenges, including greenhouse gas (GHG) emissions (Chang et al., 2017), nitrogen (N) and phosphorus (P) emissions (Herath and Satoh, 2015), as well as energy (Badiola et al., 2018), freshwater (Pahlow et al., 2015) and land use (Diana, 2009; Froehlich et al., 2018). These impacts have been studied by numerous researchers through various indicators. For example, MacLeod et al. (2020) quantified greenhouse gas (GHG) emissions in the context of global aquaculture; Pahlow et al. (2015) revealed that substituting fish meal and fish oil with terrestrial feed ingredients could escalate stress on freshwater resources; Badiola et al. (2018) provided an overview of energy utilization in recirculating aquaculture systems (RAS); Herath and Satoh (2015) conducted an analysis of the environmental repercussions stemming from phosphorus and nitrogen in aquaculture. Furthermore, aquaculture can give rise to considerable environmental repercussions on other ecosystems and human well-being, encompassing issues like species invasion (Cuthbert et al., 2021), alterations in biodiversity (Haubrock et al., 2021; Su et al., 2021), depletion of mangrove habitats (Hamilton, 2013), and the utilization of fish-derived resources. The employment of fish meal and fish oil in aquaculture feed has garnered substantial attention, as excessive usage could pose a threat to the populations of wild juvenile fish and marine ecosystems (Cao et al., 2015; Gephart et al., 2021).

Concerning aquaculture sustainability, several studies have undertaken comprehensive analyses. For example, Guzmán-Luna et al. (2021) evaluated the sustainability of tilapia aquaculture, scrutinizing water, energy, and land footprints. Their findings suggested that replacing livestock with tilapia did not yield heightened sustainability. Hallström et al. (2019) and Bianchi et al. (2022) assessed seafood sustainability by amalgamating climate and nutritional aspects. Their investigations indicated that embracing a broader range of seafood choices, such as pelagic species like sprat, herring, and mackerel, could enhance both health and climate benefits. Moreover, consuming farmed bivalves (e.g., mussels and oysters) demonstrated the highest nutrient benefits with minimal emissions. Shepon et al. (2021) delved into sustainable aquaculture development using a nutrition-sensitive approach, revealing that no single system could seamlessly achieve a nutrition-environment win-win scenario. Regarding aquaculture production systems, Dong et al. (2022) evaluated the sustainability of ten major aquaculture production systems (APSs) in China. Additionally, Stentiford et al. (2020) introduced a framework to delineate, monitor, and mitigate potential adverse impacts of escalated production while considering interactions with land-based food systems.

China is the world's largest aquaculture producer, accounting for over 60% of the world's production, and the per capita fish consumption in China was about 35.1 kg in 2010 (Han et al., 2018). Since the economic reforms in the late 1970s, China has experienced rapid

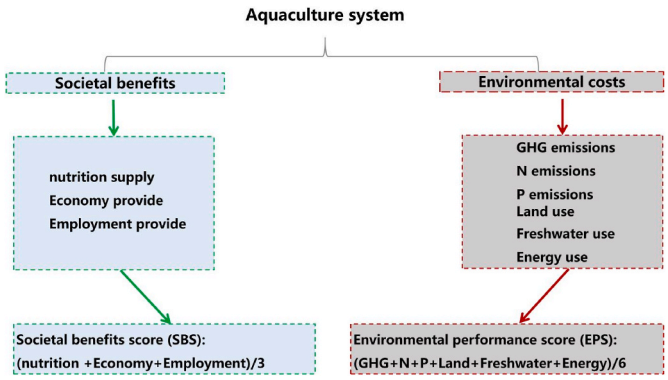


Fig. 1. Conceptual framework of two sub-systems (aquaculture benefits and aquaculture costs). Societal benefits account for nutrition supply but also for economic development and employment; environmental performance is measured through greenhouse gas, nitrogen and phosphorus releases, land use, energy, and freshwater use.

aquaculture development, with production increasing from 6.5 Mt in 1991 to 52.2 Mt in 2020, at an average annual growth rate of 23.44% (China Fishery Statistical Yearbook, 1991; 2021). The array of species has witnessed remarkable expansion, evolving from a mere few dozen species in the 1950s to over 400 species presently. Notably, fish production has consistently held a dominant stature (Gui et al., 2018). Owing to diverse aquaculture environments, significant disparities in cultivated species are apparent across various geographical regions. For instance, freshwater fish stand as the principal cultivated species in freshwater aquaculture, whereas shellfish and algae take center stage in seawater aquaculture (Gui et al., 2018). Furthermore, the aquaculture model is perpetually in flux, with intensified farming gaining momentum (Chen and Gao, 2023). Nevertheless, as aquaculture production continues its upward trajectory, its environmental performance undergoes alteration. Despite these changes, comprehensive investigations into benefits and costs remain wanting. Many studies have narrowly focused on specific indicators of benefit or cost through life cycle assessment. While Dong et al. (2022) have established an evaluation system that integrates social, economic, environmental and resource criteria, and optimizes aquaculture systems through a holistic approach, their research was based on specific criteria rather than an analysis at a higher organizational level. To address this gap, we analyze performance of Chinese aquaculture from the perspective of two sub-systems: societal benefits and environmental performance (Fig. 1). Leveraging individual benefits and costs, this approach amalgamates the advantages and environmental performance linked to aquaculture by establishing distinct sub-systems. Through the quantification of aquaculture sustainability, it provides a more robust framework for evaluating the strides made in sustainable development within the aquaculture sector.

This study aims to provide a comprehensive analysis of the societal benefits and environmental performance of Chinese aquaculture by examining the performance of each indicator in each province, comparing the benefits and costs performance of each province, and comparing the difference of marine and freshwater aquaculture. The results of this study could help to optimize the development of adequate strategies for maximizing productivity while minimizing negative impacts, and provide a scientific framework for other countries as aquaculture continues to expand globally (Jiang et al., 2022).

2. Materials and methods

2.1. Definition and conceptual model

In this study, two sub-systems was defined, which include societal benefits and environmental performance (Fig. 1). This framework allowed us to assess the sustainability of aquaculture production in

China by minimizing costs, while ensuring adequate nutrition supply, economic and employment efficiency. Important issues from each subsystem were selected based on Gephart et al.'s (2021) and Dong et al.'s (2022) identification. These issues include societal benefits such as nutrition, economy, and employment; environmental performance such as land use, freshwater use, energy use, GHG emissions, N and P emissions.

2.2. Societal benefits

This study included 49 species that account for the majority of production in Chinese aquaculture (Table S1). For each species, we derived content densities (mg/per 100 g) of ten essential nutrients or micronutrient from the Food Nutrition Facts Inquiry Platform (http://nlc.chinanutri.cn/fq/), established by the China Center for Disease Control and Prevention, and the Chinese Nutrition Society. These nutrients included calcium, iron, zinc, phosphorus, protein, magnesium, vitamin E, selenium, monounsaturated fatty acids (M-fatty acid), and polyunsaturated fatty acids (P-fatty acids). For species without corresponding nutrient densities in the Food Nutrition Facts Inquiry Platform, mean nutrient densities from the same taxonomic order were used as a proxy. The composite nutrient score for each province ($S_{nu,p}$) is calculated as follows:

$$S_{nu,p} = \sum_{j=1}^{n=10} S_{j,p} \tag{1}$$

Where $S_{j,p}$ denotes the standardized nutrient intensity of nutrition composition j for province p. The $S_{j,p}$ is calculated as follows:

$$S_{j,p} = \frac{S'_{j,p} - S'_{j,p,\min}}{S'_{j,p,\max} - S'_{j,p,\min}} \tag{2}$$

Where $S'_{j,p,min}$ and $S'_{j,p,max}$ are the minimum and maximum values of province p in the nutrition composition j, respectively. $S_{j,p}$ is obtained through standardization of raw data, where raw nutritional intensity ($S'_{j,p}$, t nutrition/t live weight production) values for each nutrition composition were rescaled from 0 to 1, whereby 1 means more nutrition can be produced per unit yield and 0 means less nutrition per unit production.

The $S'_{j,p}$ for each nutrient of each province was calculated:

$$S'_{j,p} = \frac{Nu_{j,p}}{P_p} \tag{3}$$

$Nu_{j,p}$ is the sum of the nutrition composition j for all of the species in the province p:

$$Nu_{j,p} = \sum_{i=1}^{m} Nu_{i,j,p} \tag{4}$$

Where $Nu_{i,j,p}$ represents the amount of nutrition composition j for species i in province p, and it calculated as follows:.

$$Nu_{i,j,p} = P_{i,p} * Per_{edible,i} * ND_{i,j} \tag{5}$$

Where $per_{edible,i}$ is the percentage of edible food for species i, $ND_{i,j}$ is nutrient densities (content per 100g) of each species for nutrition composition j.

For the aquaculture gross product and population number in the aquaculture sector, the data was extracted from the China Fishery Statistics Yearbook (2021) (Table S2). Since aquaculture gross product and population number are total values for aquaculture, the total production for each province was used when calculating the economy intensity and employment intensity.

2.3. Environmental performance

2.3.1. Land use

In this study, land use refers to the land area allocated to the growth of a unit of production, which applies to aquaculture systems that are factory-farming and pond-based. The land area used per unit of aquaculture production was calculated, and the production and land area data were extracted from the China Fishery Statistics Yearbook (2021) (Table S2). To calculate land use for factory-farming, a ratio of 1:10 (water area of tanks to land use for the RAS farm) was applied according to Dong et al. (2022).

$$LU_p = \frac{A_{po} + A_F}{P_p} \tag{6}$$

Where A_{po} and A_F represent areas of pond and factory-farming, P_p is the production (production from pond and factory) for province p.

2.3.2. Freshwater use

According to Pahlow et al. (2015), the freshwater use was calculated for feed-related method. which quantifies the amount of water consumed during the production and usage of aquaculture feed. For the water footprint, which can be divided into green water, blue water and the gray water (Hoekstra et al., 2011). The amount of commercial feed is calculated as follows:

$$P_{feed} = FCR_i * P_i \tag{7}$$

Where P_{feed} is the total amount of aqua-feed consumed by species i, FCR_i is the feed conversion ratio (tonne of feed/tonne of live weight) for species i, P_i is the production (tonnes) of i.

The amount of specific feed ingredient used per species is determined as:

$$Feed_{i,k} = f_{i,k} * P_{feed} \tag{8}$$

Where $Feed_{i,k}$ is the annual amount of feed ingredient k in ton/year fed to species i, and $f_{i,k}$ is the fraction of feed ingredient k in the composition of the commercial feed applied to species i. The water footprint related to commercial WF_{feed} (m^3/tonne live weight) is determined for each species i as:

$$WF_{feed,i} = \sum_{k=1}^{l} Feed_{i,k} * WF_{i,k} \tag{9}$$

Where $WF_{feed,i}$ is the green, blue and gray water footprint of species i, and $WF_{i,k}$ is the water footprint of feed ingredient k (Pahlow et al., 2015), and l the number of feed ingredients.

Therefore, the freshwater use in a province ($WF_{feed,p}$) is a combination of the use of all species.

$$WF_{feed,p} = \sum_{i=1}^{m} WF_{feed,i} \tag{10}$$

So the freshwater use intensity for province p is calculated as follow:

$$EI_{WF,p} = \frac{WF_{feed,p}}{P_p} \tag{11}$$

2.3.3. Energy use

Electricity is used on farms primarily for pumping water, aerator, and the amount of energy was calculated based on pond areas and farm areas (Boyd et al., 2007; Xu et al., 2007). The energy use intensity ($EI_{e,p}$) is calculated as follows:

$$EI_{e,p} = \frac{KW \cdot h_{pp} + KW \cdot h_{pf} + KW \cdot h_{ap} + KW \cdot h_{af}}{P_p} \tag{12}$$

Where $KW \cdot h_{pp}$, $KW \cdot h_{pf}$, $KW \cdot h_{ap}$ and $KW \cdot h_{af}$ represent energy use for

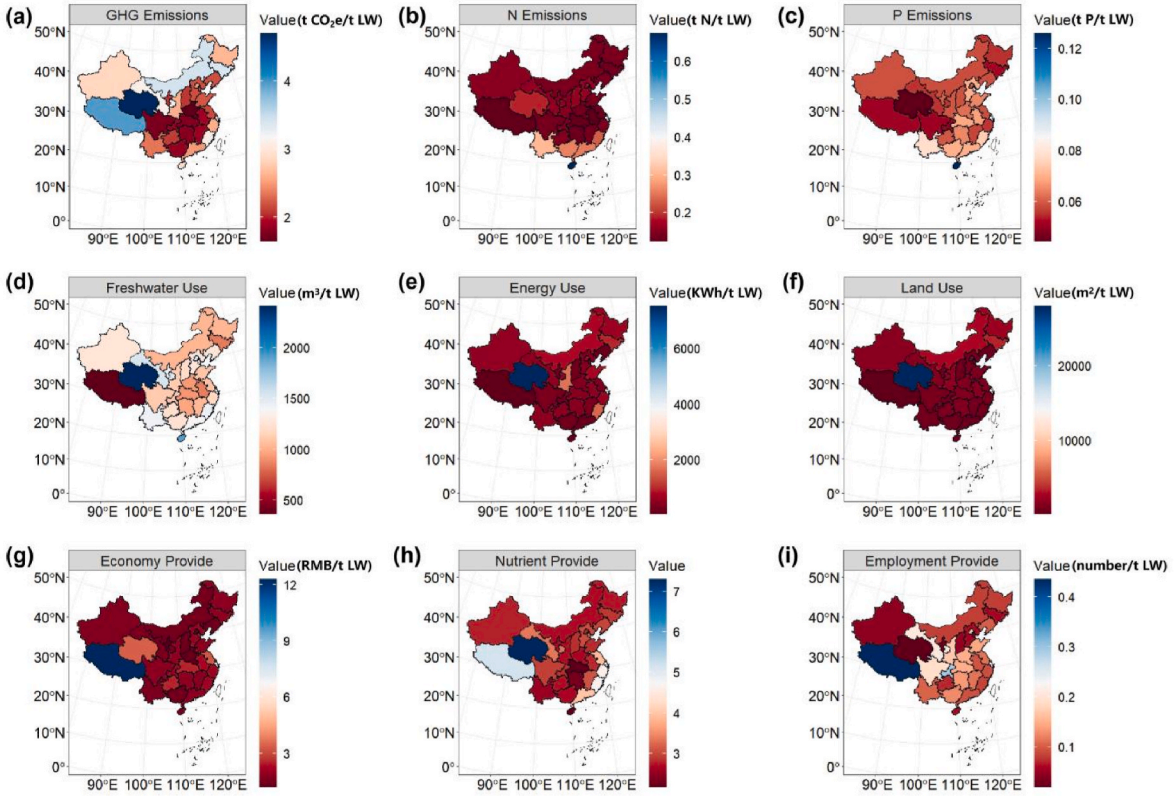


Fig. 2. The nine indicators values of each province for freshwater aquaculture. (a)–(f) Six indicators of aquaculture costs (greenhouse gas, nitrogen and phosphorus emissions intensity, freshwater, energy and land use intensity); (g)–(i) three indicators of aquaculture benefits (economy, nutrient and employment intensity); LW: Live weight; Shanghai, Hong Kong, Macao and Taiwan provinces are not included in this study due to the data limitations; there are no unit due to normalization for nutrient provide.

pumping water in pond, pumping water in factory, aerator in pond and aerator in factory; P_p represent the production (pond and factory) for each province.

2.3.4. Greenhouse gas emissions

GHG emissions of aquaculture mainly include N_2O, CO_2, and CH_4, and an increase in GHG emissions can cause a series of environmental problems (Doney et al., 2012). The emissions intensity of the main aquaculture groups (fish, shrimps, crabs, bivalves, etc.) were adapted from Xu et al. (2022), who considered GHG emissions from feed materials, on-farm energy use, aquatic N_2O and CH_4. The greenhouse gas emissions intensity is calculated as follows:

$$EI_{GHG,p} = \frac{GE_{feed,p} + GE_{energy,p} + GE_{N_2O,p} + GE_{fertilizer,p}}{P_p} \quad (13)$$

Where $GE_{feed,p}$, $GE_{energy,p}$, $GE_{N2O,p}$ and $GE_{fertilizer,p}$ represented GHG emissions from feed, energy, N_2O and fertilizer for province p respectively.

2.3.5. Nitrogen emissions

Nitrogen was estimated as the difference between the N content of each feed component & fertilizer and the species-specific N contents of aquaculture species (Gephart et al., 2021), and the formula for N emissions intensity ($EI_{N,p}$) for each province as follows:

$$N_{emission,i} = \sum_{k=1}^{l} Feed_{i,k} * N_k + N_{fertilizer,i} - N_i \quad (14)$$

$$EI_{N,p} = \frac{\sum_{i=1}^{m} N_{emission,i}}{P_p} \quad (15)$$

Where N_k $N_{fertilizer,i}$ and N_i represent the nitrogen content of feed component k, the nitrogen content of fertilizer and the nitrogen content of species i respectively, and l represented number of feed component.

2.3.6. Phosphorus emissions

Phosphorus emissions were also calculated at the on-farm stage, using the method outlined in Huang et al. (2020). Nitrogen was estimated as the difference between the input and the species-specific N contents of aquaculture species. The N content of each feed component was calculated according to Robb et al. (2017), with N content coming from crude protein. Nitrogen content of aquatic products was obtained from Luo et al. (2018), with species being divided into marine and freshwater categories. The total input was estimated using the following formula:

$$Ph_{inp,i} = Ph_{hat,i} / PUE_i \quad (16)$$

Where $P_{hat,i}$ represent the phosphorus content of species i, PUE_i represent the proportion of supplementary P applied to feed species and fertilize the aquatic ecosystems that are recovered in harvested fish. The parameters of PUE for each species were driven by Huang et al. (2020).

The emission intensity ($EI_{Ph,p}$) for province p is as follows:

$$EI_{Ph,p} = \frac{\sum_{i=1}^{m} Ph_{inp,i} - \sum_{i=1}^{m} Ph_{hat,i}}{P_p} \quad (17)$$

2.4. Sustainable score of each province

To calculate the composite societal benefits and environmental performance scores for each province and ensure comparability of indicators, the raw indicator values were re-scaled from 0 to 100, whereby

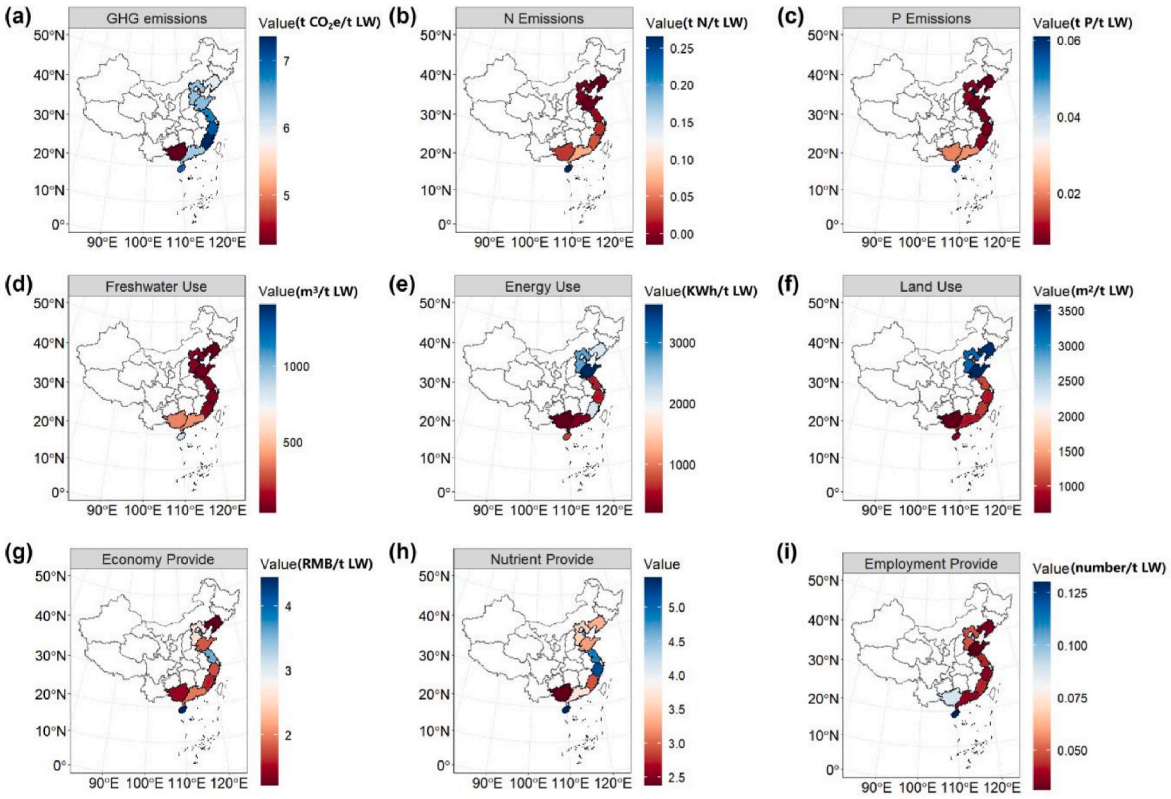


Fig. 3. The nine indicators values of each province for mariculture. (a)–(f) Six indicators of aquaculture costs (greenhouse gas, nitrogen & phosphorus emissions intensity and freshwater, energy & land use intensity); (g)–(i) three indicators of aquaculture benefits (economy intensity, nutrient intensity and employment intensity); Shanghai, Hong Kong, Macao and Taiwan provinces are not included in this study due to the data limitations; there are no unit due to normalization for nutrient provide.

100 means the highest benefits or costs and 0 means the lowest benefits and costs among nine indicator. The raw indicator intensity values are normalized by the following formulas:

$$S_t = \frac{S'_{t,p} - S_{t,p,\min}}{S_{t,p,\max} - S_{t,p,\min}} * 100 \quad (18)$$

Where $S'_{t,p,min}$ and $S'_{t,p,max}$ are the minimum and maximum values of province p for the indicator t, respectively.

3. Results

3.1. Performance of nine indicators for each province

When only considering freshwater aquaculture, Qinghai province performed the worst in terms of GHG emissions, freshwater use, energy use, and land use, with values of 4.68 t CO_2e/t live weight, 2402.61 m^3/t live weight, 7496.72 TJ/t live weight, and 27868.85 m^2/t live weight, respectively (Fig. 2). However, it had the smallest values for P emissions. On the other hand, Hainan province had the largest values for N and P emissions, with values of 0.67 t N/t live weight and 0.13 t P/t live weight, respectively, compared to other provinces (Fig. 2). Shaanxi and Fujian provinces performed poorly in terms of energy use, while Jilin and Inner Mongolia had high values for land use compared to most other provinces. In contrast, Tibet province performed the best in terms of freshwater use, energy use, and land use intensity, with values of 222.22 m^3/t live weight, 59.78 TJ/t live weight, and 363.22 m^2/t live weight, respectively (Fig. 2). For aquaculture benefits, Qinghai province had the highest nutrient intensity with a value of 7.29, while Tibet province had the highest economy and employment intensity, with values of 12.27 ten thousands RMB/t live weight and 0.43 numbers/t live weight, respectively (Fig. 2g and h). Conversely, Ningxia, Hubei, and Qinghai provinces had the lowest economy, nutrient, and employment intensity, with values of 1.18 ten thousands RMB/t live weight, 2.16, and 0.02 numbers/t live weight, respectively (Fig. 2g, h & i).

There were significant differences in aquaculture costs among provinces for mariculture (Fig. 3). Shandong province had the highest values for land and energy use, but it had lower P emissions intensity than most other provinces. Meanwhile, Tianjin and Zhejiang provinces had the worst performance in terms of freshwater use and P emissions intensity (1399.99 m^3/t live weight and 0.06 t P/t live weight). Fujian and Hainan provinces had the highest intensity of GHG and N emissions (7.35 t CO_2e/t live weight and 0.26 t N/t live weight), respectively. On the other hand, Guangxi province had the smallest values for land and energy use (621.01 m^2/t live weight and 205.94TJ//t live weight) and GHG emissions (4.26 t CO_2e/t live weight). Additionally, Liaoning province performed the best in terms of freshwater use and N emissions (34.08 m^2/t live weight and −0.02N/t live weight). In terms of aquaculture benefits, Hainan province had the highest nutrient, economy, and employment intensity. Conversely, Guangxi, Liaoning, and Shandong provinces had the lowest values for nutrient, economy, and employment intensity.

When comparing composite aquaculture, the values of the nine indicators in 10 coastal provinces showed varying degrees of changes (Fig. 4). It was observed that the values of land and energy use indicators decreased for most provinces, with Liaoning province exhibiting the great reduction. The values of land and energy use for Liaoning province decreased from 3518.34 m^2/t live weight and 2133.61 TJ/t live weight to 1429.87 m^2/t live weight and 846.68/t live weight, respectively. However, in contrast, the values of N emissions increased for all provinces, with Hainan province experiencing the largest change, increasing from 0.26 t N/t live weight to 0.54 t N/t live weight. Furthermore, for freshwater use and P emissions, the values of most provinces also

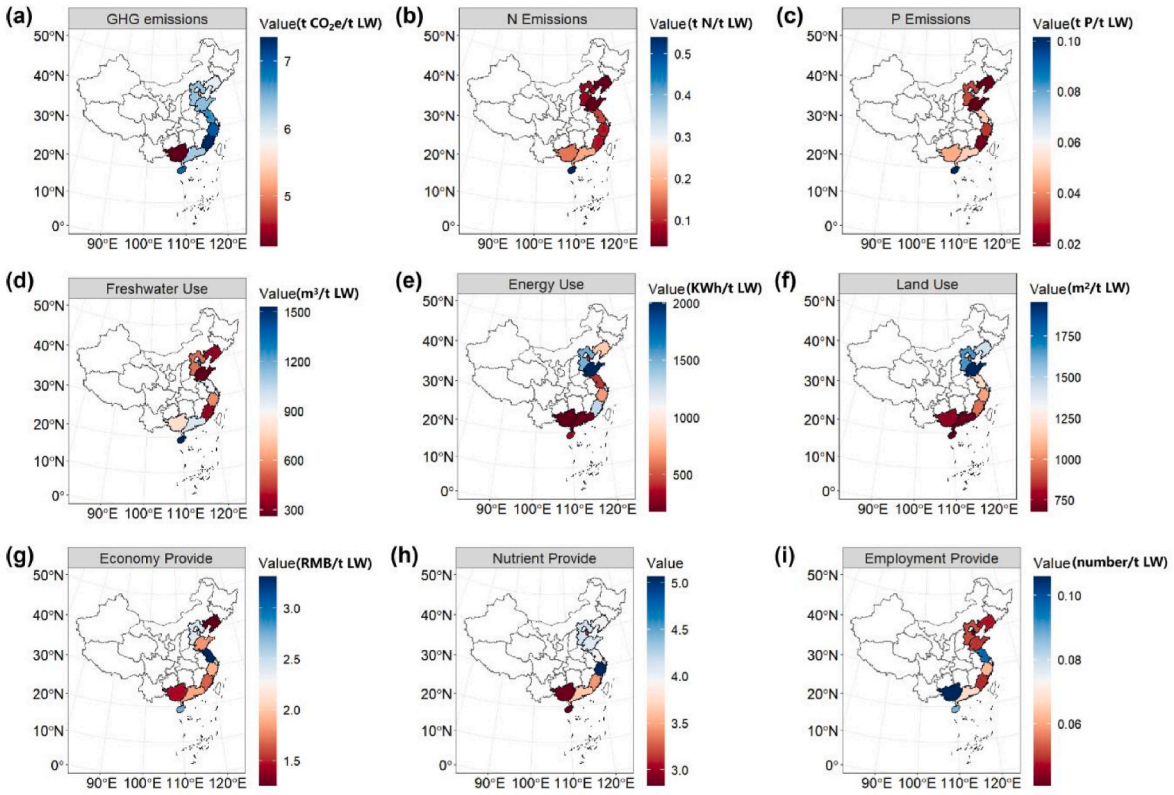


Fig. 4. The nine indicators values of each province for composite aquaculture (include freshwater and mariculture) of 10 costal provinces. (a)–(f) Six indicators of aquaculture costs (greenhouse gas, nitrogen & phosphorus emissions intensity and freshwater, energy & land use intensity); (g)–(i) three indicators of aquaculture benefits (economy intensity, nutrient intensity and employment intensity); Shanghai, Hong Kong, Macao and Taiwan provinces are not included in this study due to the data limitations; there are no unit due to normalization for nutrient provide.

increased when considering both freshwater and mariculture together. Employment intensity also increased for most provinces, except for Tianjin and Hainan provinces. Regarding nutrition and economy indicators, nearly half of the provinces showed an increase while the other half showed a decrease. Hainan province had the largest value change for nutrition and economy indicators, which decreased by 2.60 (no unit due to normalization) and 1.67 ten thousands RMB/t live weight, respectively (Figs. 3 and 4).

3.2. Benefits and costs of Chinese aquaculture

China is divided geographically into seven regions: Northeast China, North China, East China, South China, Central China, Northwest China, and Southwest China (Fig. 5). Based on the average score of each province within each region, it was observed that the Southwest region had the highest average score (30.65) for societal benefits, while Northeast China performed the worst (Fig. 5a). For environmental performance, South China and Northwest China had the highest scores (38.55 and 33.91, respectively), whereas Central China performed the best with the lowest score (22.54) (Fig. 5c). There were significant differences among provinces in the Northwest and Southwest regions compared to other regions (Fig. 5b and d). For instance, in Northwest China, Qinghai province had the highest score (39.88), while Ningxia province had the lowest score (4.13) for societal benefits, resulting in a difference of 35.75. In contrast, the provinces in Central China had smaller differences for aquaculture benefits compared to other regions. The provinces in South China had the largest differences for aquaculture costs compared to the other six regions, while the three provinces in Central China had the smallest difference.

In terms of freshwater aquaculture, there were 12 provinces with a benefits-to-costs ratio of more than 1, with Tibet having the highest ratio of 6.09. Chongqing and Sichuan provinces also had high ratio of 2.85 and 2.13, respectively. On the other hand, Hainan province had the lowest ratio of 0.07 compared to other provinces in freshwater aquaculture (Fig. 6a). For mariculture, More than half of the provinces have a ratio greater than 1. Jiangsu and Guangxi provinces performed the best (with ratio scores of 2.71 and 2.05, respectively) among the 10 provinces. By contrast, Liaoning and Shandong provinces had the smallest ratio scores of 0.33 and 0.34, respectively (Fig. 6b). When freshwater and mariculture were considered simultaneously, there were only three provinces with ratio scores exceeding 1, with Tibet province performing the best (with a ratio score of 4.86) among the 30 provinces. Ningxia province had the smallest ratio score of 0.17 (Fig. 6c).

4. Discussion

This study provides a comprehensive analysis of the composite aquaculture benefits and costs for each province in China based on nine indicators, bridging the gaps of previous studies that mainly focused on assessing either the benefits or environmental impacts of aquaculture or a limited geographic region. (Cao et al., 2007; Wu et al., 2018; Zheng et al., 2019; Yang et al., 2020; Xu et al., 2023).

4.1. Benefits and environmental performance of provinces and regions

This study revealed that the performance of Chinese aquaculture cannot be assessed accurately by relying on a single environmental impact or nutritional indicator since the nine indicators evaluated varied significantly across different provinces. These differences can be attributed to variations in farming modes and species cultured in each province. Feed and fertilizer use account for the differences in aquaculture costs across provinces. For example, in Qinghai province, where

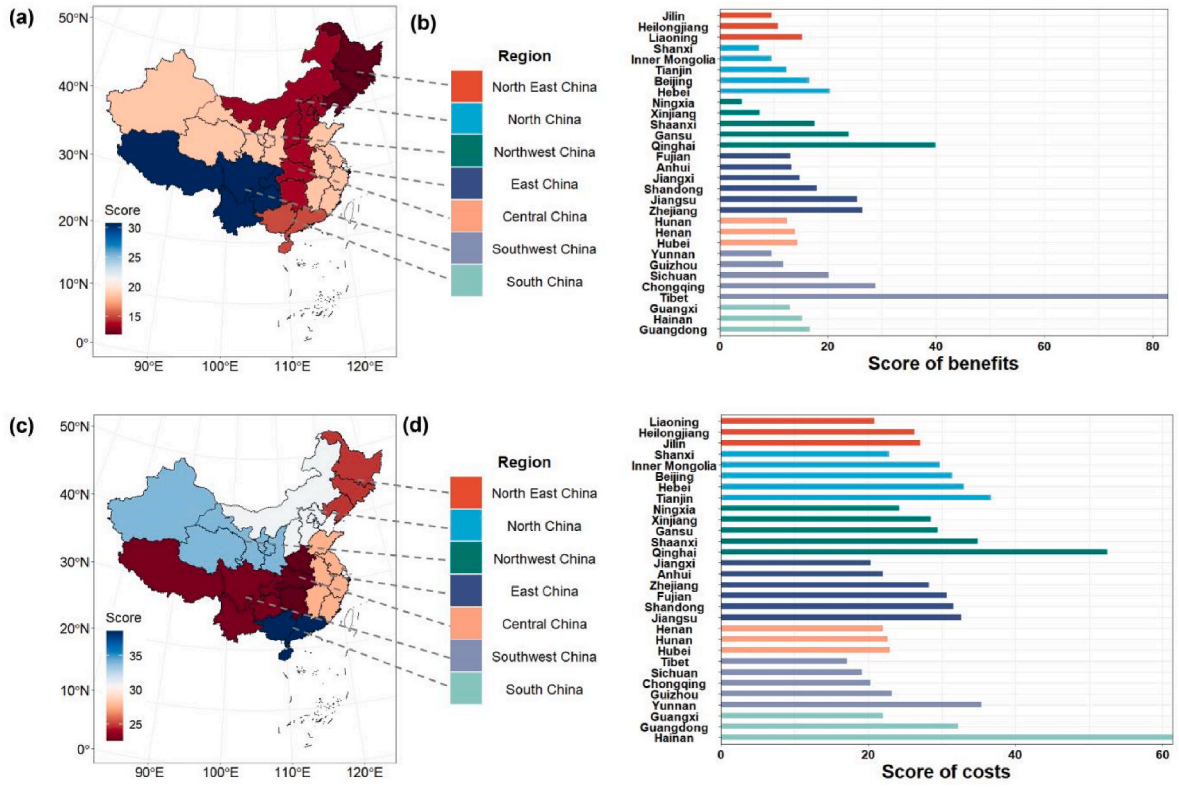


Fig. 5. Sustainability scores of the two sub-systems for seven regions for composite aquaculture (include freshwater and mariculture). (a) The mean benefits score for each region; (b) the histograms represent the benefits score for each province in each region; (c) the mean costs score for each region; (d) the histograms represent the costs score for each province in each region. Shanghai, Hong Kong, Macao and Taiwan provinces are not included in this study due to the data limitations.

trout (*Squaliobarbus ourriculus*) accounts for around 97.71% of the 49 species production, the freshwater intensity (which was calculated based on the method of Pahlow et al. (2015) from feed use) of trout is higher than that of most other species, leading to high freshwater use intensity. The significant prevalence of trout aquaculture in Qinghai province can primarily be attributed to the region's cold-water-oriented aquaculture environment, well-developed technologies, substantial market demand, and the hallmark of excellent water quality. These factors collectively provide a conducive milieu for the cultivation of trout (Li et al., 2021). Conversely, in Hainan province, where tilapia is the main species accounting for 92.57% of total production, the province ranks poorly in nitrogen and phosphorus emissions due to tilapia's emission intensity, which is the second and fifth highest among the 49 species studied using the calculation method of Gephart et al. (2021) and Huang et al. (2020) respectively. As far back as 2006, the Hainan provincial government had already promulgated the government document titled "Tilapia Industrialization Action Plan in Hainan Province" (Hainan Provincial Government Office). This initiative underscored that the climatic conditions, water resources, and farming techniques in Hainan province served as the bedrock for the progressive expansion of tilapia cultivation. This strategic approach has consequently led to a substantial share of tilapia production in Hainan province. Moreover, tilapia has low nutrient value (Food Nutrition Facts Inquiry Platform) and economic value (Dong et al., 2022), contributing to Hainan province's worst societal benefits performance in freshwater aquaculture.

The difference in cultured species also contributes to the variations in aquaculture costs among coastal provinces. In Hainan and Tianjin provinces, the largest N emissions intensity and P emissions intensity were recorded, respectively. This is because grouper and marine white shrimp (*Litopenaeus Vannamei*), which are among the most cultured species in these provinces, accounted for about 80.77% of the total production (China Fishery Statistics Yearbook, 2021). For instance, marine white shrimp alone contributed 98.37% of all the output in Tianjin province. The relatively elevated production percentage of marine white shrimp can be attributed to the significant backing provided by governmental support. Through collaborative endeavors between fishery management and science and technology promotion departments, the maturation of breeding technology has reached a commendable level in Tianjin province (Xie and Sun, 2021). On the other hand, Liaoning province had the smallest N and P emissions intensity due to the higher percentage of bivalves among the 10 provinces. Because bivalves not only do not consume nitrogen from feed, but can also fix nitrogen during their growth by shells and muscles (Jansen and van den Bogaart, 2020). Therefore, the higher the proportion of bivalve production, the lower the value of feed-related aquaculture cost indicators. Certain shellfish species exhibit more stringent temperature requirements. For instance, scallops thrive in cooler water temperatures, contributing to a larger share of cultured shellfish production in provinces like Liaoning and other northern regions (Guo and Luo, 2016). Jiangsu and Guangxi provinces performed well in mariculture according to benefits to costs ratio, but their performance differed among the nine indicators. After the normalization of all indicators, the land and energy use and GHG emissions of Guangxi province were all zero, but its nutrition and economic indicators were also low. In contrast, Jiangsu province had not any zero performance on some indicators within costs and benefits. Tibet province had the largest economic value, which was attributed to the high percentage (85.00%) of production of mandarin fish, the species with the highest economic benefits among the 49 cultured species (Zhu et al., 2022; Chen et al., 2023). South China recorded the highest environmental performance score, as indicated by the average composite aquaculture value, mainly due to the species cultured. Tilapia not only remains the dominant species in freshwater aquaculture but also greatly surpasses bivalve production, which only accounts for 2.97% of mariculture. This disparity in production levels contributes to the high environmental performance. Tibet province

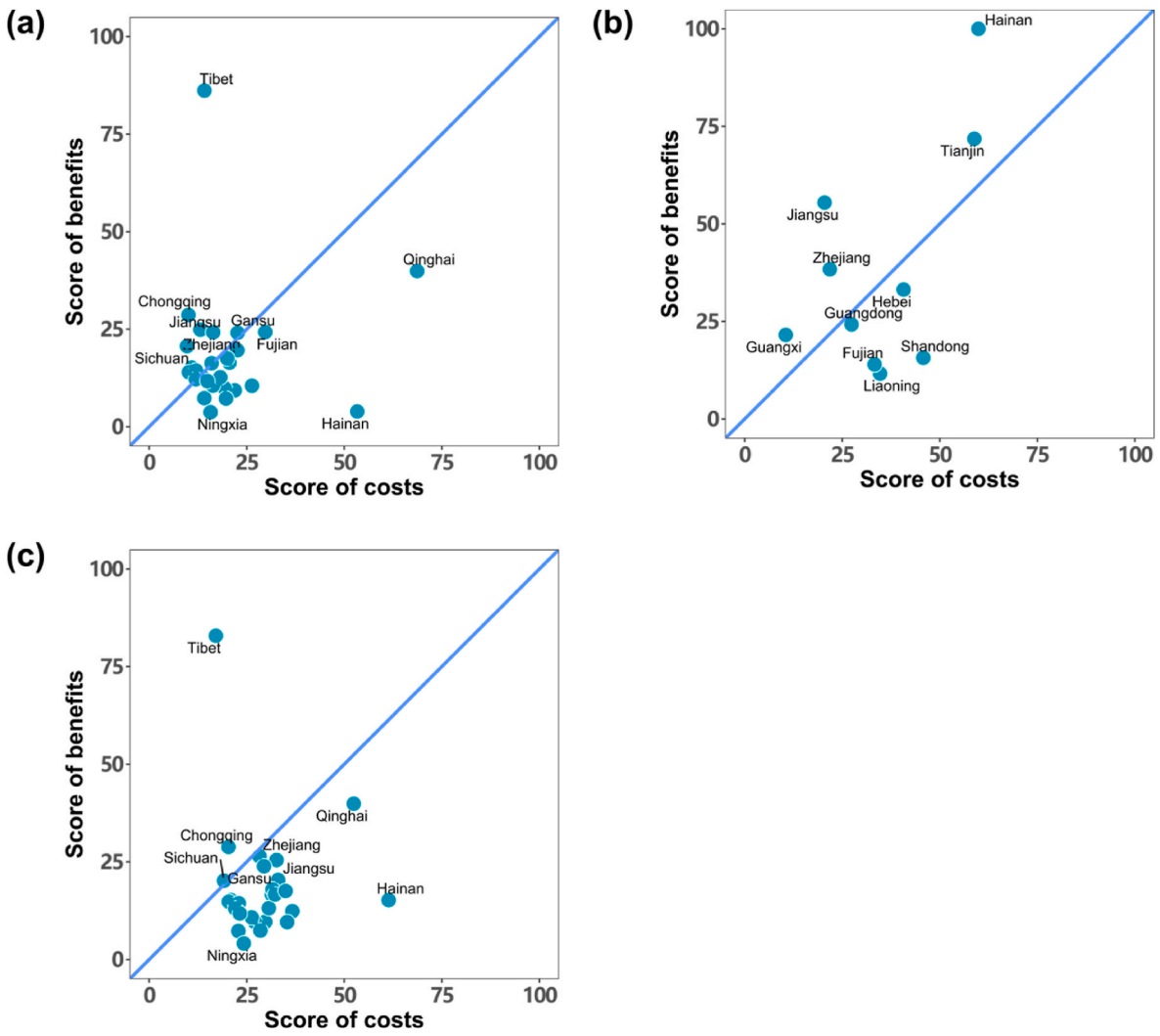


Fig. 6. Benefits-to-costs ratio of each province. (a) Freshwater aquaculture for each province; (b) mariculture for each province; (c) composite aquaculture (include freshwater and mariculture) for each province. The blue line [Ratio (SS) = 1] represents the score of benefits is equal to the score of costs. Shanghai, Hong Kong, Macao and Taiwan provinces are not included in this study due to the data limitations. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

performed well in economy, nutrient, and employment intensity, contributing to the most societal benefits among the 30 provinces. In provinces with a low degree of economic development, aquaculture technology is relatively backward, requiring more people to produce aquatic products of the same yield compared with high degree of economic development provinces. In contrast, fewer people are needed to produce the same aquatic products in more developed provinces (Jiang et al., 2022).

4.2. Limitations and perspectives

Although this study provides a comprehensive analysis of Chinese aquaculture by utilizing nine indicators and two sub-systems, there are certain limitations in the results due to data limitations. Firstly, the estimation of land and energy use was based on pond and factory, which could have underestimated the actual values. Additionally, when calculating the P emissions intensity, the total production was used, and no-feed production was not excluded. However the percentage of no-feed production is small, it may have small impacts on the result. Furthermore, the indicators and two sub-systems were assumed to be equal, and several other indicators with high costs or benefits, such as biological invasion, habitat destruction, chemical pollution, were not included in this study (Cole et al., 2009; Rico et al., 2012; Ahmed and Thompson, 2019; Diagne et al., 2021). Therefore, further research incorporating these indicators could lead to more comprehensive results.

In the future, the allocation of weight between indicators and sub-systems should be established through a scientific approach (Dong et al., 2022). Given that various indicators carry differing degrees of importance in terms of environmental impacts or services, relying solely on average calculations among the indicators might obscure significant issues from being adequately highlighted. Secondly, the improvement of aquaculture technology or modes and use cleaner energy can also significantly improve the sustainability of aquaculture. For instance, the transition from extensive aquaculture systems to recirculating systems holds significant potential for energy reduction, achievable through enhancements to airlift and bio-filter designs (d'Orbcastel et al., 2009). The substitution of coal with hydro, nuclear, or natural gas can lead to substantial reductions in greenhouse gas emissions (Cao et al., 2011). Moreover, the judicious selection of highly sustainable species for aquaculture can markedly enhance the overall sustainability of aquaculture within specific regions. For instance, an evaluation of seafood nutritional diversity alongside climate impacts establishes farmed bivalves as the most sustainable species (Bianchi et al., 2022). Conversely,

prudent choices exclude species like tilapia when more sustainable alternatives are available. Finally, the feed use is the main source of many environmental impacts (such as GHG, N and P emissions). So reducing the feed conversion ratio and improving feed composition is essential for reducing the costs (Ali et al., 2010; Ansari et al., 2021). For instance, the utilization of microencapsulated algae as feed presents a dual advantage. This approach not only curbs resource consumption but also mitigates greenhouse gas emissions in commercial bivalve aquaculture (Willer et al., 2020). Besides, insect larva, single cell proteins also continue to appear cost-effective (Boyd et al., 2020). Last but not least, in order to promote the development of aquaculture in an environmentally friendly direction, more indicators should be included in the aquaculture sustainability assessment through weighting, especially for the biological invasion, biodiversity change, habitat impacts, chemical pollution and other indicators with greater environmental impact (Cranford et al., 2006; Obasohan et al., 2009; Cuthbert et al., 2021). The framework in this study will facilitate increasing benefits of aquaculture and decrease costs, and this framework would be improvement by more data collection that require interaction across government departments and a broad range of aquaculture stakeholders.

5. Conclusion

This study provides a comprehensive analysis of Chinese aquaculture industry by examining multiple indicators and considering their benefits and costs. The findings reveal that the performance of nine indicators varied significantly across the 30 provinces. Specifically, the Southwest and South regions showed the highest benefits ($SBS = 30.65$) and costs ($EPS = 38.55$) respectively based on the average scores of the two subsystems, while Tibet and Hainan provinces had the highest scores for aquaculture benefits ($SBS = 82.90$) and aquaculture costs ($EPS = 61.37$) respectively, at the province level. Based on the benefits to costs ratio, Tibet province demonstrated the best performance in freshwater and composite aquaculture (6.09 and 4.86 respectively), while Jiangsu province performed the best in mariculture (2.71). To achieve greater sustainability in the future, the study suggests using recirculating system, using cleaner energy, reducing the feed conversion ratio by algal or other protein and cultured sustainable species in Chinese aquaculture industry. Besides, this study emphasizes the need to include accurate data more indicators through weighting to achieve a more accurate assessment of aquaculture sustainability. By doing so, Chinese aquaculture industry can work towards achieving greater sustainability assessment, which will have positive impacts on the environment and society improvement.

CRediT authorship contribution statement

Congjun Xu: Conceptualization, Methodology, Formal analysis, Data curation, Writing – original draft, Visualization. **Guohuan Su:** Conceptualization, Methodology, Formal analysis, Data curation, Writing – review & editing, Supervision, Funding acquisition. **Kangshun Zhao:** Methodology, Formal analysis, Writing – review & editing. **Xianghong Kong:** Formal analysis, Writing – review & editing. **Hongxia Wang:** Methodology, Formal analysis. **Xiaoqi Xu:** Methodology, Writing – review & editing. **Ziqi Li:** Methodology, Formal analysis. **Min Zhang:** Conceptualization, Methodology, Formal analysis, Data curation, Writing – review & editing, Supervision. **Jun Xu:** Conceptualization, Methodology, Data curation, Writing – review & editing, Project administration, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2023.138645.

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